

Response surface methodology for predicting quality characteristics of beef patties added with flaxseed and tomato paste.

Response surface methodology was used to study the effect of flaxseed flour (FS) and tomato paste (TP) addition, from 0 to 10% and 0 to 20% respectively, on beef patty quality characteristics. The assessed quality characteristics were color (L, a, and b), pH and texture profile analysis (TPA). Also, sensory analysis was performed for the assessment of color, juiciness, firmness, and general acceptance. FS addition reduced L and a values and decreased weight loss of cooked products ($P<0.05$). An opposite effect was observed when TP was added ($P<0.05$). All TPA parameters decreased when percentages of FS and TP were increased in the formulation of beef patties. Furthermore, FS and TP addition adversely affected the sensory characteristics of the cooked product ($P<0.05$); nevertheless, all sensory characteristics evaluated had an acceptable score (>5.6). Thus FS and TP are ingredients that can be used in beef patty preparation. Copyright © 2014 Elsevier Ltd. All rights reserved.